

Velocities From Distance vs. Time graphs – Average Velocity and Instantaneous Velocity

Synopsis: Most real motions have acceleration. In fact, things never stay at rest or move at a constant speed in a straight line in perpetuity. We considered a freefalling motion as an example of an accelerated motion. We found that with a constant acceleration ($a = g = -9.8 \text{ m/s}^2$), the velocity increases linearly with time. This, in turn, causes the distance covered by the falling object to increase over the same time interval. This Note teaches how to think about “velocity” when it is constantly changing. A velocity at one instant of

time is different from a velocity at another instant of time. Remember the equation, $v_f = v_i + at$, where v_f is the final velocity (m/s), v_i is the initial velocity (m/s), a is the acceleration (m/s^2), and t is the time (s)? This is the equation that tells you *how* the velocity changes from one instant to the next – this is it. You should note that when $a = 0$, the final velocity is the same as the initial for all times (why?) – a constant speed motion with the velocity vs. time graph having a flat line with zero slope (why? The answers should be obvious). You should also see from this equation that the plot of v_f vs. time is a straight line with a (nonzero) constant slope, the slope being the acceleration. (Can you explain this using your knowledge of a linear equation?). Now, because of this nonzero acceleration in this equation of v_f , you get the curved distance vs. time graph. This may not be obvious yet: The clue is the area under the curve of velocity vs. time graph. That is fine. What we focus on in this Note is that using the curved distance vs. time graph, we

can learn about 1) The “average velocity” over the whole time ($v_{\text{average}} = \bar{v} = \frac{d_{\text{total}}}{t}$), and 2) The “Instantaneous velocity,” the velocity at a specific instant of time, different from one moment to the next ($v_{\text{instantaneous}} = v_f = v_i + at$).

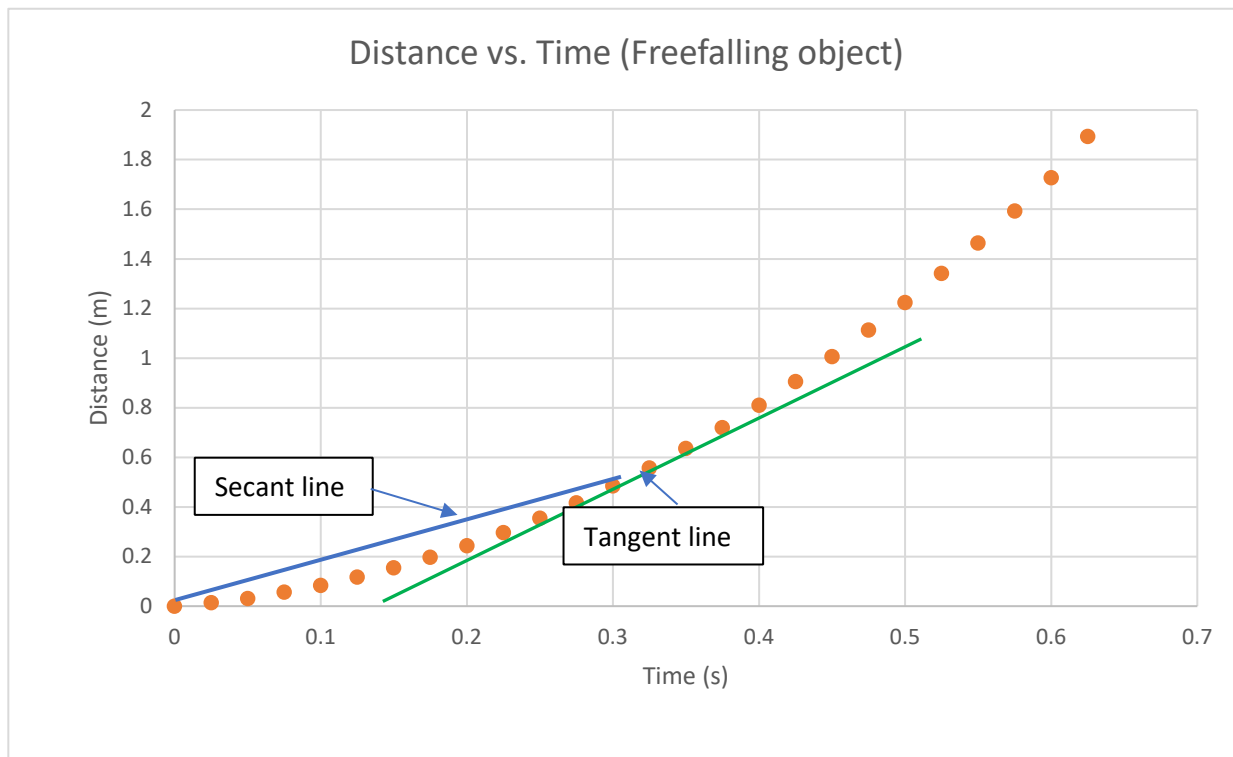
I will introduce two types of lines: Secant and Tangent. Secant connects two data points used for average velocity, and tangent passes through a single data point used for instantaneous velocity. The velocity is the slope of the respective line.

Performance Expectations: Define average velocity and instantaneous velocity. Extract the average velocity using the secant line and instantaneous velocity from the tangent line in the distance vs. time graph.

Preamble

- 1) Why is the slope of velocity vs. time graph zero in the constant speed motion?
- 2) Using your knowledge of linear equations, explain why the plot of v_f as a function of time has the slope as the acceleration.

Instructions



- 1) The above is the distance vs. time graph of the freefalling object. At $t = 0.3$ s, we can calculate the average and instantaneous velocity using the secant and tangent lines, respectively.

Q1) Describe how the secant and tangent lines are drawn (one sentence each).

Q2) Are the secant and tangent line slopes the same? _____ Which is steeper? _____ Estimate the average and the instantaneous velocity at $t = 0.3$ s.

Average velocity =

Instantaneous velocity =

- 2) Estimate the average and instantaneous velocity at $t = 0.6$ s by drawing a secant to and tangent line at $t = 0.6$ s.

Average velocity =

Instantaneous velocity =

- 3) Examine the velocities at 0.3 and 0.6 s. The average and instantaneous velocities are connected, i.e., a numerical relationship exists. What do you think that is? (Hint: the numerical relationship is simple.)

In Calculus Notation

Average velocity over an interval is a difference quotient (secant slope): $v_{\text{avg}} = \Delta x / \Delta t$. Instantaneous velocity is what that quotient approaches as the interval shrinks to zero — the derivative (tangent slope):

$$v(t) = dx/dt = \lim (\Delta t \rightarrow 0) \Delta x / \Delta t$$

In your own words, explain why shrinking the secant's interval makes average velocity converge to instantaneous velocity: